Worksheet: The Surplus From a Sale

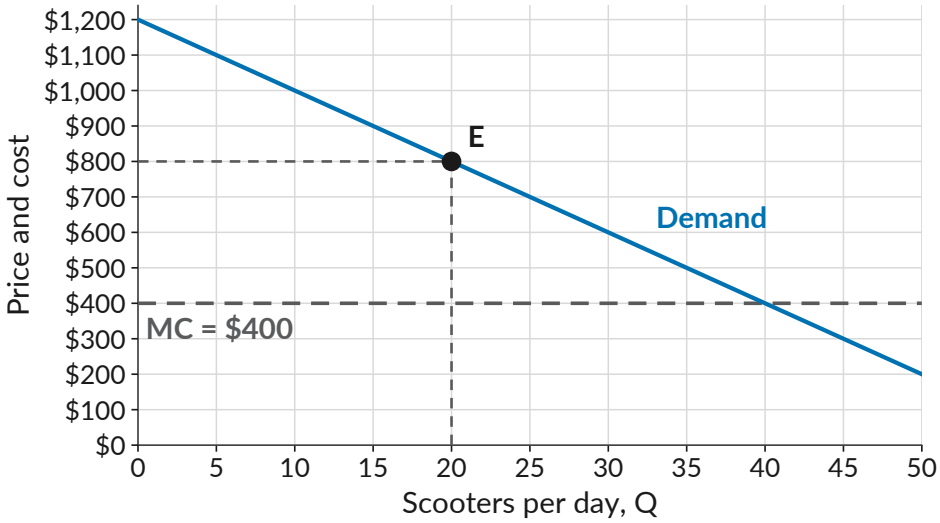
ECON 201

Definitions

- **Notation**: WTP is the buyer's willingness to pay, P is the price, and MC is marginal cost.
- **Gains from one sale**: the buyer gains WTP $-$ P and the firm gains $P -$ MC. The **joint surplus** is their sum, WTP $-$ MC.
- **Consumer surplus**: the sum of the buyers' gains, the area between the demand curve and the price.
- **Producer surplus**: the sum of the firm's gains, the area between the price and marginal cost.
- **Areas**: a triangle is $\frac{1}{2} \times$ base $\times$ height, and a rectangle is base $\times$ height.

Electric scooters

A company sells electric scooters. The graph shows the demand curve for its scooters and its marginal cost. It maximizes profit at the point E, where it's $MR = MC$.



1. How many scooters does the company sell, and at what price?

2. How much is the 10th buyer willing to pay? What does this buyer gain from the sale, and what does the company gain?

3. What is the joint surplus from this sale?

4. Shade consumer surplus and producer surplus on the graph, and find the value of each.

5. Why does the company capture the larger share of the surplus?